Sports Forecasting with Language Models

Generative Modeling, Natural Language Processing

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1 Problem Description

We propose a language-model framework for improving probabilistic forecasts of NBA game outcomes. Specifically, we ask whether training an open-weight language model on long-context pregame text (press conferences, official injury reports, beat-reporter coverage, and analyst previews) improves forecasts over frontier models given the same context but not trained for this task.

Our project adapts the merger-arbitrage forecasting framework of Jajal et al. [1].

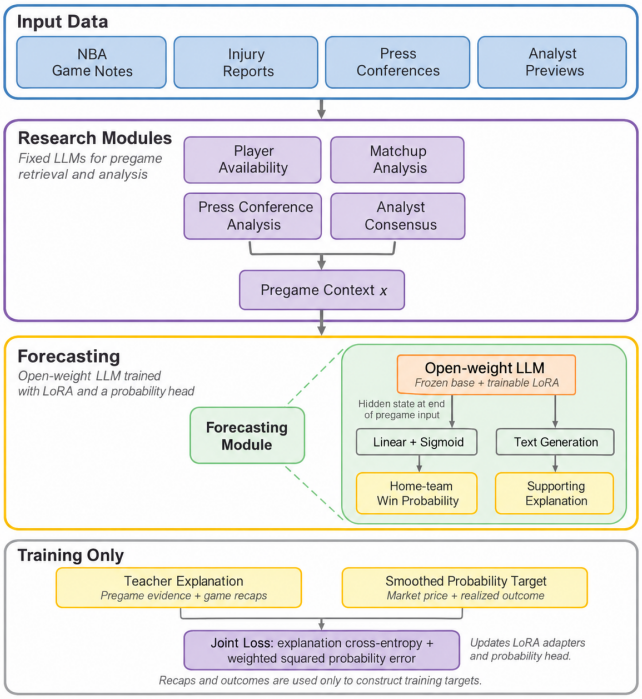


Figure 1: Proposed NBA forecasting architecture, adapted from Jajal et al. [1]. Research modules organize pregame evidence for a forecasting LLM trained using hindsight-guided supervision.

2 Challenges

The main challenge will be both obtaining structured pregame data to train the model and matching historical predictions to point-in-time Polymarket baselines. We must also prevent information

leakage from edited web pages and from the model already knowing past outcomes, so we will test only on games after every evaluated model's knowledge cutoff.

3 Dataset

We plan to combine pregame team notes from official NBA/team websites, press-conference transcripts from team media channels, and analyst consensus from published previews and forecasts, along with structured team statistics, game outcomes, and historical Polymarket prices. To obtain consecutive historical coverage, we may request archives from the NBA or third-party data providers. We will split games chronologically into 70% training, 15% validation, and 15% hold-out.

4 Preprocessing

The beauty of LLMs is that they allow us to use unstructured game notes, transcripts, and analyst commentary without extensive manual feature engineering. For each game, we keep only documents timestamped before a fixed pre-game time, and group the text into modules (injuries, matchup, press conferences, analyst consensus).

5 Learning Method

We will adapt the framework of Jajal et al. [1], replacing merger-specific research with basketball-focused modules. These modules will organize pregame evidence into context for a fine-tuned language model, which outputs a win probability and a supporting explanation.

Following Jajal et al. [1], training targets pair hindsight-guided explanations, written by a teacher model using game recaps, with probabilities that blend the Polymarket price and the realized outcome. Using an existing supervised fine-tuning implementation on an open-weights model with LoRA, we will train the forecasting model on these targets. Beyond Jajal et al. [1], we add a probability head on the model's final hidden state and train jointly,

$$\mathcal{L}(\theta) = -\sum_{t} \log p_\theta(z_t \mid x, z_{<t}) + \lambda \, (\hat{p}_\theta(x) - p^*)^2,$$

where x is pregame context, z is the target explanation, $\hat{p}_\theta$ is the head's win probability, and p^* is the smoothed target. Token-level cross-entropy rewards imitating probability strings, whereas the Brier term directly rewards calibrated forecasts.

6 Evaluation

We will evaluate forecasts on chronologically held-out games using the Brier score, $\frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$, where p_i is the predicted home-team win probability and $y_i \in \{0, 1\}$ is the outcome; lower scores indicate better probabilistic predictions. Our primary baselines will be frontier models given identical pregame context but not fine-tuned for this task, with Polymarket prices recorded at the same pregame cutoff as a market reference. We will ablate the probability head against standard supervised fine-tuning and test differences with a paired bootstrap over games. Success would be a statistically supported improvement over frontier models, demonstrating the value of domain-specific fine-tuning.

References

[1] Hinal Jajal, Michal Mucha, Charles Sweat, Chris Pulman, Charlie Flanagan, and Peter Anderson. Global merger-arbitrage forecasting with language models. *arXiv preprint arXiv:2607.09921*, 2026. URL `https://arxiv.org/abs/2607.09921`.